

Structure of on Finite Extensions of $\mathbb{Q}_p$

In this section we establish a structure result for a finite field extension K of $\mathbb{Q}_p$ which is very similar to the structure of $\mathbb{Q}_p$ itself.

The main result is the following:

Proposition: There is an absolute value $|\cdot|_p : K \rightarrow \mathbb{R}^{\geq 0}$ which satisfies:

1. $|a|_p$ is the usual absolute value for $a \in \mathbb{Q}_p$.
2. $|\alpha|_p = 0$ if and only if $\alpha = 0$.
3. $|\alpha \cdot \beta|_p = |\alpha|_p |\beta|_p$.
4. $|\alpha + \beta|_p = \max\{|\alpha|_p, |\beta|_p\}$.
5. The image of $|\cdot|_p : K \setminus \{0\} \rightarrow \mathbb{R}^{>0}$ is discrete.

This has a number of consequences, which we shall also prove:

- The subset $R = \{\alpha : |\alpha|_p \leq 1\}$ which is a subring of K consists of all those elements of K which are *integral* over $\mathbb{Z}_p$. (Recall that α is integral over $\mathbb{Z}_p$ if it satisfies a monic polynomial with coefficients in $\mathbb{Z}_p$.)
- There is an element $\pi \in R$ such that every element of K is of the form $\alpha = u \cdot \pi^k$ for $u \in R$ a unit and k an integer such that $|\alpha|_p = |\pi|_p^k$. In particular, there is a positive integer e such that $p = u_\pi \cdot \pi^e$ where u_π is a unit in R . It follows that $|\pi|_p = p^{-1/e}$.
- The quotient $R/\langle \pi \rangle$ is a finite field extension of $\mathbb{F}_p = \mathbb{Z}/\langle p \rangle$. In particular, it has p^f elements for some positive integer f .
- We have the formula $[K : \mathbb{Q}_p] = ef$.

In order to prove these results, we will prove some properties of the field norm $\text{Nm}_{K/\mathbb{Q}_p}$ which we recall below.

Field Norm and trace

Given a field extension K of k which is a finite dimensional vector space over k where $[K : k] = n$ denotes the vector space dimension.

Norm. We define the norm $\text{Nm}_{K/k} : K \rightarrow k$ as follows. Choose a basis $S = \{\alpha_1, \dots, \alpha_n\}$ of K as a vector space over k . This gives a k -algebra homomorphism $M_S : K \rightarrow M_n(k)$ where the latter is the algebra of $n \times n$ matrices with entries in k . This satisfies $M_S(\alpha) = (a_{i,j})$ where

$$\alpha \cdot \alpha_i = \sum_{j=1}^n a_{i,j} \alpha_j$$

We then define $\text{Nm}_{K/k}(\alpha) = \det(M_S(\alpha))$ which is independent of the choice of basis S . It is also clear that this satisfies:

- $\text{Nm}_{K/k}(\alpha \cdot \beta) = \text{Nm}_{K/k}(\alpha) \cdot \text{Nm}_{K/k}(\beta)$.

- $\text{Nm}_{K/k}(a) = a^{[K:k]}$ for every element $a \in k$.

More generally, we have:

Exercise: Given a subfield L of K which contains k , and $\alpha \in L$, show that

$$\text{Nm}_{K/k}(\alpha) = (\text{Nm}_{L/k})^{[K:L]}$$

(Hint: We can combine bases $U = \{\beta_1, \dots, \beta_p\}$ of L over k and $T = \{\gamma_1, \dots, \gamma_q\}$ of K over L into the basis $S = \{\beta_i \cdot \gamma_j\}$ of K over k . The matrix $M_S(\alpha)$ is then a block-diagonal matrix with diagonal blocks $M_U(\alpha)$.)

Note: A more intricate argument can be used to show that for $\alpha \in K$, we have

$$\text{Nm}_{K/k}(\alpha) = \text{Nm}_{L/k}(\text{Nm}_{K/L}(\alpha))$$

but we do not need this in the discussion below.

Trace. We define the trace $\text{Tr}_{K/k} : K \rightarrow k$ as $\text{Tr}_{K/k}(\alpha) = \text{Trace}(M_S(\alpha))$ which is again independent of the choice of basis.

It is clear that this satisfies

$$\text{Tr}_{K/k}(a \cdot \alpha + \beta) = a \cdot \text{Tr}_{K/k}(\alpha) + \text{Tr}_{K/k}(\beta)$$

for $a \in k$ and $\alpha, \beta \in K$. We also see that $\text{Tr}_{K/k}(a) = n \cdot a$ for $a \in k$.

In particular, when $n \neq 0$ in k (note that k *could* be of finite characteristic!) we see that $\text{Tr}_{K/k}(1) \neq 0$. It follows that the trace pairing

$$(\alpha, \beta)_{K/k} = \text{Tr}_{K/k}(\alpha \cdot \beta)$$

is non-degenerate in this case. This is a non-degenerate bilinear symmetric pairing on the k -vector space K . In particular, this is the case for $k = \mathbb{Q}_p$.

Definition of $|\cdot|_p$

The main result of this note is:

Given a finite field extension K of $\mathbb{Q}_p$, the function

$$|\alpha|_{p,K} := |\text{Nm}_{K/\mathbb{Q}_p}(\alpha)|_p^{1/[K:\mathbb{Q}_p]}$$

is an absolute value on K .

Here the $|\cdot|_p$ on the right is the usual p -adic absolute value of $\text{Nm}_{K/\mathbb{Q}_p}(\alpha)$ which is an element of $\mathbb{Q}_p$.

(The following proof is based on an argument given in the book on p -adic numbers by N. Koblitz.)

Some properties of this function follow immediately from the above discussion:

- The meaning of $|a|_{p,K} = |a|_p$ is unchanged for $a \in \mathbb{Q}_p$.
 - More generally, if L is a subfield of K , and α is an element of L , then $|\alpha|_{p,K} = |\alpha|_{p,L}$.
- Thus, we can drop the subscript K in the notation and just use $|\alpha|_p$ for an α which lies in *some* finite extension of $\mathbb{Q}_p$.
- $|\alpha|_p \geq 0$ and equality holds if and only if $\alpha = 0$.
 - The function is multiplicative: $|\alpha \cdot \beta|_p = |\alpha|_p |\beta|_p$.
 - The image of $K \setminus \{0\}$ is contained in the discrete subset $\{p^{k/n} : r \in \mathbb{Z}\}$ of $\mathbb{R}^{>0}$.

The main result thus reduces to proving the inequality:

$$|\alpha + \beta|_p \leq \max\{|\alpha|_p, |\beta|_p\}$$

Now, this inequality is obvious when α or β is 0. Thus, without loss of generality, we may assume that $|\beta|_p \geq |\alpha|_p > 0$. In this case, we put $\gamma = \alpha/\beta$ so that $|\gamma|_p \leq 1$. Dividing the above inequality by $|\beta|_p$, we see that we are reduced to proving that:

$$|1 + \gamma|_p \leq 1 \text{ whenever } |\gamma|_p \leq 1$$

Using the results above, we can assume that $K = \mathbb{Q}_p[\gamma]$ so that K is spanned by $S = \{1, \gamma, \dots, \gamma^{n-1}\}$ where n is the degree of the minimal polynomial of γ over $\mathbb{Q}_p$.

Comparison with a metric

For a matrix $A \in M_n(\mathbb{Q}_p)$ let us define

$$\|A\| = \max_{i,j} |a_{i,j}|_p$$

and note that this satisfies the following:

- $\|A\| \geq 0$ and equality holds if and only if $A = 0$.
- $\|a \cdot A\| = |a|_p \|A\|$ for $a \in \mathbb{Q}_p$.
- $\|A + B\| \leq \max\{\|A\|, \|B\|\}$ for $B \in M_n(\mathbb{Q}_p)$.

Exercise: Given a matrix $A = (a_{i,j})$ in $M_n(\mathbb{Q}_p)$ show that

$$|\det(A)|_p \leq \left(\max_{i,j} |a_{i,j}|_p \right)^n = \|A\|^n$$

(Hint: Note that the determinant of A is a sum of terms of the form $\pm (a_{1,\sigma(1)} \cdots a_{n,\sigma(n)})$ for some permutation σ of $\{1, \dots, n\}$. Also use the inequality $|a + b|_p \leq \max\{|a|_p, |b|_p\}$ for $a, b \in \mathbb{Q}_p$.)

- $|\det(A)|_p \leq \|A\|^n$ as seen above.

Let $A = M_S(\gamma)$ be the associated $n \times n$ matrix with entries in $\mathbb{Q}_p$. The given condition $|\gamma|_p \leq 1$ is the same as $|\det(A)|_p \leq 1$.

Claim: There is a uniform bound $C \geq 1$ for the entries of A^i for all positive integers i . In other words, $\|A^i\| \leq C$ for all positive integers i .

Note that $M_S((1 + \gamma)^m) = (1 + A)^m$. Thus, assuming the claim we have:

$$|1 + \gamma|_p^m \leq |(1 + \gamma)^m|_p \leq \|(1 + A)^m\| \leq \left| \binom{m}{i} \right|_p \|A^i\| \leq C$$

using the fact that the p -adic absolute value of any integer is at most 1.

Taking m -th roots, we get $|1 + \gamma|_p \leq C^{1/m}$ for all m . It follows that $|1 + \gamma|_p \leq 1$ as required.

Proof of the Claim

The proof is by contradiction.

We are given that $A = M_S(\gamma)$ satisfies $|\det(A)|_p \leq 1$.

Suppose that for each positive integer j , there is an i_j so that $\|A^{i_j}\| > j$.

There is an entry $b_j \in \mathbb{Q}_p$ of the matrix A^{i_j} such that $j < |b_j|_p = \|A^{i_j}\|$. Putting $B_j = A^{i_j}/b_j$, we have $\|B_j\| = 1$.

This means that the entries of B_j lie in $\mathbb{Z}_p$. Since $\mathbb{Z}_p$ is *compact*, there is a subsequence j_k for which the matrices B_{j_k} converge in $M_n(\mathbb{Z}_p)$ to some matrix B . By continuity we see that $\|B\| = 1$. Furthermore, since $A \cdot B_j = A^{1+i_j}/b_j = B_j \cdot A$, we see, by continuity, that $A \cdot B = B \cdot A$.

On the other hand, we have

$$|\det(B_j)|_p = \frac{|\det(A^{i_j})|_p}{|b_j^n|_p} < \frac{|\det(A)^{i_j}|_p}{j^n} \leq \frac{1}{j^n}$$

which shows that $\det(B) = 0$.

Thus, there is a non-zero element λ in the vector space K such that $B\lambda = 0$. Since K is a field, $\{\lambda, \gamma\lambda, \dots, \gamma^{n-1}\lambda\}$ is *also* a basis of K .

Note that $A^i\lambda = \gamma^i\lambda$ since $A = M_S(\gamma)$. This means $B\gamma^i\lambda = BA^i\lambda = A^iB\lambda = 0$. In other words, B must be the 0 matrix. This contradicts the assertion above that $\|B\| = 1$.

R is integral over $\mathbb{Z}_p$

We define the subset $R = \{\alpha : |\alpha|_p \leq 1\}$ in K .

By using the additive and multiplicative properties of $|\cdot|_p$, we see that R is a subring and contains $\mathbb{Z}_p$.

The characteristic polynomial of $\alpha \in K$ is

$$f(X) = \text{ch}_{K/\mathbb{Q}_p}(X) = \det(XI_n - M_S(\alpha))$$

It follows that for $a \in \mathbb{Z}_p$, we have

$$f(a) = \det(M_S(a) - M_S(\alpha)) = \text{Nm}_{K/\mathbb{Q}_p}(a - \alpha)$$

Now $|\text{Nm}_{K/\mathbb{Q}_p}(a - \alpha)|_p = |a - \alpha|_p^{[K:\mathbb{Q}_p]}$, and if we further restrict $\alpha \in R$, we have

$$|a - \alpha|_p \leq \max\{|a|_p, |\alpha|_p\} \leq 1$$

We conclude that $|f(a)|_p \leq 1$ which means that $f(a) \in \mathbb{Z}_p$ since $f(a) \in \mathbb{Q}_p$.

Exercise: Suppose $f(X) \in \mathbb{Q}_p[X]$ is such that $f(a) \in \mathbb{Z}_p$ for every $a \in \mathbb{Z}_p$. Show that $f(X) \in \mathbb{Z}_p[X]$. (Hint: Assume the contrary that some coefficient of $f(X)$ is in $\mathbb{Q}_p \setminus \mathbb{Z}_p$ and show that there is an $a \in \mathbb{Z}_p$ such that $f(a) \notin \mathbb{Z}_p$.)

It follows that α is integral over $\mathbb{Z}_p$ as required.

Note that when α lies in R , the trace $\text{Tr}_{K/\mathbb{Q}_p}(\alpha)$ is the coefficient of X^{n-1} (up to sign) in $f(X)$. Thus, it is in $\mathbb{Z}_p$. This will be useful below.

Units in R

Note that if $u \in R$ is a unit, then there is a $v \in R$ such that $u \cdot v = 1$. From $|u|_p \leq 1$ and $|v|_p \leq 1$ and $|u \cdot v|_p = 1$, we see that $|u|_p = |v|_p = 1$.

Conversely, if $u \in R$ is such that $|u|_p = 1$, then $u \neq 0$ and $v = 1/u \in K$ satisfies $|v|_p = 1$ so it also lies in R . Thus, u is a unit in R .

Existence of π

Since the image of $R \setminus \{0\}$ under $|\cdot|_p$ is discrete in $\mathbb{R}^{>0}$, there is an element π in R such that $|\pi|_p$ is the largest value *less than* 1. Note that $0 < |p|_p = p^{-1} < 1$, so $p \in R$ and

$$0 < |p|_p \leq |\pi|_p < 1$$

For any element $\alpha \in K$ we have

$$\{|\alpha \cdot \pi^{-k}|_p : k \in \mathbb{Z}\} = |\alpha|_p \cdot \{|\pi|^{-k} : k \in \mathbb{Z}\}$$

By the discreteness of this set, there is a k such that $|\alpha \cdot \pi^{-k}|_p$ is closest to 1 from below. If $|\alpha \cdot \pi^{-k}|_p \leq |\pi|_p < 1$ then $|\alpha \cdot \pi^{-k-1}|_p \leq 1$ is even closer to 1. Thus, by the definition of π , we must have $|\alpha \cdot \pi^{-k}|_p = 1$.

It follows that $u = \alpha \cdot \pi^{-k}$ is a unit in R and $\alpha = u \cdot \pi^k$ as required.

Note that for every $\alpha \in K$, the integer k is such that $|\alpha|_p = |\pi|_p^k$.

In particular, we have a positive integer e such that $p^{-1} = |p|_p = |\pi|_p^e$ so that $|\pi|_p = p^{-1/e}$.

Further note that if τ is a unit multiple of π then it has the same properties as π in the above discussion. Any such element is called a *uniformizer* of R and K .

Basis of R over $\mathbb{Z}_p$

Since K is a finite field extension of $\mathbb{Q}_p$ we have a basis $\alpha_1, \dots, \alpha_n$ of K as a vector space over $\mathbb{Q}_p$.

There is a positive integer m such that $|\alpha_i|_p \leq |\pi|_p^{-m} = p^{m/e}$ for all i . Replacing m by a larger number we see that there is a positive integer k such that $p^k \cdot \alpha_i$ is in R for all i . It follows that R contains a basis of K over $\mathbb{Q}_p$. By change of notation, let us assume that $\alpha_1, \dots, \alpha_n$ is such a basis so that R contains the $\mathbb{Z}_p$ -linear span of this $\mathbb{Q}_p$ -basis.

Since the trace pairing on K is non-degenerate, there is a dual basis $\eta_1, \dots, \eta_n$ in K such that

$$(\alpha_i, \eta_j)_{K/\mathbb{Q}_p} = \text{Tr}_{K/\mathbb{Q}_p}(\alpha_i \cdot \eta_j) = \begin{cases} 1 & \text{when } i = j \\ 0 & \text{otherwise} \end{cases}$$

Then for each $\alpha \in K$ we have

$$\alpha = \sum_{i=1}^n (\alpha, \alpha_i)_{K/\mathbb{Q}_p} \eta_i$$

is the expression of α in the basis $\eta_1, \dots, \eta_n$.

Since $\alpha_i \in R$, if $\alpha \in R$, then $\alpha \cdot \alpha_i \in R$. So $\text{Tr}_{K/\mathbb{Q}_p}(\alpha \cdot \alpha_i)$ lies in $\mathbb{Z}_p$. This shows that R is contained in the $\mathbb{Z}_p$ -linear span of $\eta_1, \dots, \eta_n$. As R is “trapped” between two free $\mathbb{Z}_p$ modules of rank n , we see that there is a basis of K over $\mathbb{Q}_p$ which is *also* a basis of R over $\mathbb{Z}_p$.

The field $F = R/\langle \pi \rangle$

It follows that the field $F = R/\langle \pi \rangle$ is finitely generated as a module over $\mathbb{Z}_p$. Since p lies in $\langle \pi \rangle$ we see that F is a finite dimensional vector space over $\mathbb{F}_p = \mathbb{Z}_p/\langle p \rangle$. Thus, F is a finite field and is isomorphic to $\mathbb{F}_q$ for some $q = p^f$.

By the results proved on finite fields, there is an element γ in F such that its minimal polynomial $\bar{g}(X)$ over $\mathbb{F}_p$ has degree precisely f . It follows that F is spanned by $1, \gamma, \dots, \gamma^{f-1}$.

Let u be an element of R that “lifts” γ . Since u is not divisible by π , we see that u is a unit.

Exercise: Check that $u^i \pi^j$ for $i = 0, \dots, f-1$ and $j = 0, \dots, e-1$ gives a basis of $R/\langle p \rangle$ which is a $\mathbb{F}_p$ -vector space.

Note that $\alpha_1, \dots, \alpha_n$ is another basis. It follows that $[K : \mathbb{Q}_p] = n = ef$

Ring generator of R over $\mathbb{Z}_p$

Let $g(X) \in \mathbb{Z}_p$ be *any* lift of $\bar{g}(X) \in \mathbb{F}_p[X]$ and $g'(X)$ be its derivative.

- The image $\bar{g}(\gamma)$ of $g(u)$ is 0 in $F = R/\langle\pi\rangle$. So $g(u)$ is a multiple of π .
- The image $\bar{g}'(\gamma)$ of $g'(u)$ is non-zero in $F = R/\langle\pi\rangle$. So $g'(u)$ is a unit in R .

Exercise: Show that either $g(u)$ is a unit multiple of π , or $g(u + \pi)$ is a unit multiple of π . (Hint: Show that $g(u + \pi) - g(u) - \pi \cdot g'(u)$ is divisible by π^2 as one does in Hensel's lemma.)

It follows that we have a unit v in R (either $v = u$ or $v = u + \pi$) which satisfies:

- v lifts γ .
- $g(v)$ is a uniformizer of R .

Exercise: Show that $1, v, \dots, v^{n-1}$ is a basis of R over $\mathbb{Z}_p$.

Let $f(X) \in \mathbb{Z}_p[X]$ be the minimal polynomial of v (which is monic). We see that R is naturally isomorphic to $\mathbb{Z}_p[X]/\langle f(X) \rangle$ with v as the image of X . Further note that $f(X) \bmod p = \bar{g}(X)^e$. Since R is a domain, $f(X)$ is irreducible.